
Improving Masked Diffusion via Confidence-Weighted Sampling and Error-Corrective Resampling

Aref Mousavi

Nima Zare

Mostafa Karbalai

Abstract

Discrete diffusion models have recently emerged as a promising non-autoregressive alternative for language modeling. However, standard sampling strategies can suffer from poor semantic integration and error propagation during generation. Building upon the Simple and Effective Masked Diffusion Language Models (MDLM) framework, this project introduces two novel sampling strategies: *Confidence-Weighted Sampling*, which prioritizes the unmasking of high-certainty tokens, and *Error-Corrective Resampling*, which allows for dynamic error correction while strictly preserving the expected diffusion noise schedule. Furthermore, we show that our proposed sampling method improves reasoning capability, as validated on the Concept-to-Sentence Generation task. Our comprehensive evaluation demonstrates that these proposed samplers noticeably reduce Generation Perplexity (GenPPL) while maintaining high lexical diversity. Additionally, we also provide mathematical justification for the token balance in our Error-Corrective Resampling scheme.

1. Introduction

The dominant paradigm in language modeling relies heavily on autoregressive (AR) generation. While highly effective, AR models are constrained by their strict left-to-right generation order, which limits flexible text manipulation and heavily constrained generation tasks. Discrete masked diffusion models present a compelling non-autoregressive alternative. These models initiate the generation process with a fully masked sequence and iteratively unmask or refine tokens over multiple timesteps.

Despite their theoretical appeal, current Masked Diffusion Language Models (MDLM) [1] face significant challenges during inference. The standard Denoising Diffusion Probabilistic Model (DDPM) discrete sampler [2] unmask tokens based purely on a predefined noise schedule, entirely ignoring the model’s predictive confidence at individual positions. This uniform scheduling can force the model to commit to highly uncertain tokens early in the generation process, leading to compounding contextual errors. Furthermore, standard discrete text diffusion lacks a mechanism to revise or “correct” previously generated tokens once they have been unmasked.

To address these critical limitations, this project proposes two enhancements to the MDLM inference phase:

- **Confidence-Weighted Sampling (CWS):** An improved unmasking rule that dynamically reallocates unmasking probability mass toward tokens with high predictive confidence, establishing a robust early context.
- **Error-Corrective Resampling (ECR):** A dynamic sampling mechanism that enables previously unmasked tokens to be remasked and regenerated. This is accompanied by a mathematical guarantee ensuring that the expected global noise schedule remains perfectly preserved.

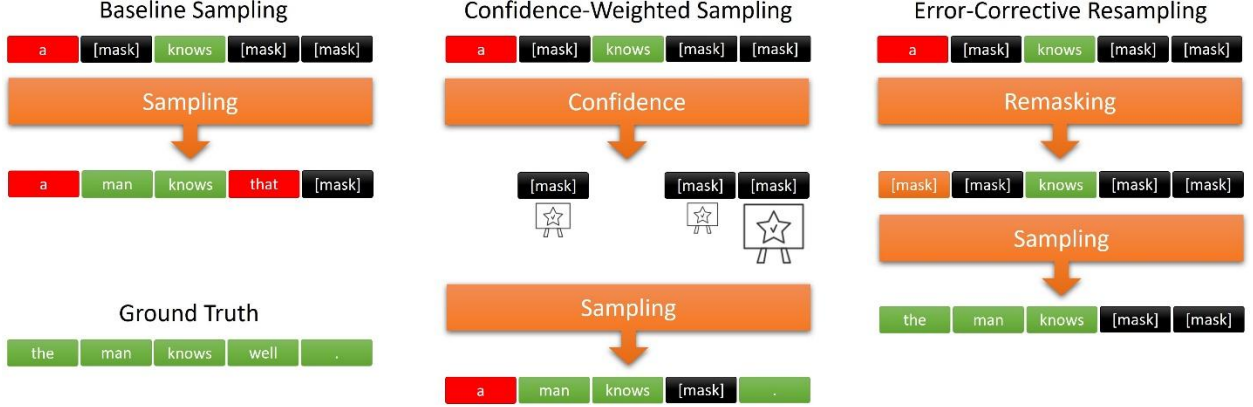


Figure 1. Comparison of sampling strategies. While the baseline propagates early errors, Confidence-Weighted Sampling prioritizes high-certainty tokens to establish context, and Error-Corrective Resampling actively fixes early mistakes to match the ground truth.

Both methods are applied entirely at inference time, requiring zero additional training or architectural modifications (see Figure 1 for a visual comparison of these sampling strategies).

To rigorously validate these sampling strategies, we conduct comprehensive evaluations across two primary paradigms. First, we assess standard unconditional text generation from a completely masked state, measuring fluency and lexical diversity through Generative Perplexity (Gen-PPL), Self-BLEU, Distinct-2, and Distinct-3 metrics. Second, we evaluate the model’s generative commonsense reasoning capabilities via the CommonGen dataset [3]. For this concept-to-sentence task, we introduce a novel constraint-based conditioning mechanism, demonstrating that our sampling enhancements significantly improve the model’s ability to seamlessly integrate required semantic constraints without degrading grammatical coherence.

2. Background: Baseline Sampling

Our baseline relies on the foundational Masked Diffusion Language Model (MDLM) framework. In this architecture, a predefined forward diffusion process dictates how input sequences are corrupted by independently transitioning clean tokens to a special `[mask]` token. The training objective for the parameterized model p_θ , is to learn the reversal of this degradation: predicting the categorical distribution of the original, uncorrupted sequence x_0 from a partially masked state at any given timestep t .

2.1 DDPM-Style Discrete Sampling

Let $x_t \in \{\mathcal{V} \cup \{[mask]\}\}^L$ represent the discrete sequence at time $t \in \{0, \dots, T\}$, where $\mathcal{V}$ is the vocabulary and L is the sequence length. We define a continuous noise schedule $\sigma(t) \geq 0$, which yields the masking probability $m(t) = 1 - e^{-\sigma(t)}$. This $m(t)$ represents the global probability that any token is masked at time t .

During the reverse generation process, we step backwards in time from a current step t to a subsequent, “cleaner” step $s = t - \Delta t$. We denote the masking probabilities at these steps as $m_t = m(t)$ and $m_s = m(s)$. Because masking decreases as we generate text, the term $(m_t - m_s)$ represents the global probability mass allocated for unmasking tokens during this specific transition.

Let $p_\theta(x_0^{(i)} = v | x_t, t)$ denote the model’s predicted probability that the original token at position i is vocabulary item v . For each sequence position $i \in \{1, \dots, L\}$, the transition from x_t to x_s operates under two rules:

1. Absorbing Unmasked Positions:

If the token at position i has already been generated (i.e., $x_t^{(i)} \neq [\text{mask}]$), it remains permanently fixed for all subsequent steps:

$$x_s^{(i)} = x_t^{(i)}$$

2. Sampling Masked Positions:

If the token is currently masked (i.e., $x_t^{(i)} = [\text{mask}]$), we construct an unnormalized categorical probability distribution $q^{(i)}(v)$ over all vocabulary tokens and the $[\text{mask}]$ token. The distribution is defined piecewise:

$$q^{(i)}(v) = \begin{cases} p_\theta(x_0^{(i)} = v | x_t, t) * (m_t - m_s) & \text{if } v \neq [\text{mask}] \\ m_s & \text{if } v = [\text{mask}] \end{cases}$$

The new token for position i at step t is then sampled from this distribution: $x_s^{(i)} \sim \text{Cat}(q^{(i)})$.

2.2 Limitation of the Baseline:

As demonstrated by the formulation of $q^{(i)}(v)$, This method determines the rate of unmasking based entirely on the global schedule difference $(m_t - m_s)$. This scalar is applied uniformly across the entire sequence. Consequently, the decision to unmask is entirely agnostic to the underlying token distribution p_θ . Furthermore, due to the absorbing nature of the transition, once a token is unmasked it remains permanently fixed for all subsequent steps. Consequently, the model possesses no mechanism to revise or correct early prediction errors during later stages of generation.

3. Methodology

To address the limitations of the MDLM, we introduce Confidence-Weighted Sampling and Error-Corrective Resampling.

3.1 Confidence-Weighted Sampling (CWS)

Uniform unmasking in early, sparse contexts forces the model to make uncertain guesses, risking compounding errors. CWS resolves this by prioritizing the model's most certain predictions. By adopting this "easy-first" approach, the model establishes a reliable foundation of highly predictable tokens, providing a robust structural context to guide the generation of more difficult tokens later.

Let $\Delta m = m_t - m_s$. We define the confidence at each masked position as:

$$c_i = \max_v p_\theta(x_0^{(i)} = v | x_t, t)$$

We introduce a confidence exponent $\beta \geq 0$ to scale this confidence: $\tilde{c}_i = c_i^\beta$. Let $\mathcal{M}_t = \{i | x_t^{(i)} = [\text{mask}]\}$ be the set of masked positions, and $M = |\mathcal{M}_t|$. The normalization factor is defined as:

$$\alpha = \frac{\Delta m * M}{\sum_{j \in \mathcal{M}_t} \tilde{c}_j}$$

For each masked position $i \in \mathcal{M}_t$, the unmasking mass is calculated as $u_i = \alpha * \tilde{c}_i$. The updated categorical probabilities for masked positions become:

$$q^{(i)}(v) = \begin{cases} p_\theta(x_0^{(i)} = v | x_t, t) * u_i & \text{if } v \neq [\text{mask}] \\ m_t - u_i & \text{if } v = [\text{mask}] \end{cases}$$

The reverse update samples $x_s^{(i)} \sim \text{Cat}(q^{(i)})$ for $i \in \mathcal{M}_t$, while currently unmasked positions remain unchanged.

3.2 Error-Corrective Resampling (ECR)

Even with an “easy-first” generation strategy, early tokens are predicted under sparse context and may become incompatible as the rest of the sequence materializes. ECR introduces a dynamic self-correction mechanism. By continuously evaluating the sequence and selectively re-hiding tokens that conflict with the newly established context, the model is granted the opportunity to regenerate them. This iterative refinement ensures global coherence and fixes early hallucinations without requiring a complete restart.

Let $\Delta m = m_t - m_s$. Given a correction factor $\gamma \geq 0$, we define the time-dependent remasking rate as $\gamma_t = \gamma * m_t$, and the per-position remasking probability as $p_{\text{remask}} = \gamma_t * \Delta m$. To compensate for remasking, we define the extra unmasking mass:

$$p_{\text{extra}} = p_{\text{remask}} * \frac{1 - m_t}{m_t}$$

This value is clipped by $p_{\text{extra}} \leftarrow \min(p_{\text{extra}}, m_s)$. For masked positions $\mathcal{M}_t = \{i | x_t^{(i)} = [\text{mask}]\}$, the update categorical is:

$$q^{(i)}(v) = \begin{cases} p_\theta(x_0^{(i)} = v | x_t, t) * (\Delta m + p_{\text{extra}}) & \text{if } v \neq [\text{mask}] \\ m_s - p_{\text{extra}} & \text{if } v = [\text{mask}] \end{cases}$$

We sample an intermediate token $\tilde{x}^{(i)} \sim \text{Cat}(q^{(i)})$. For unmasked positions ($i \notin \mathcal{M}_t$), we sample a remasking indicator $r_i \sim \text{Bernoulli}(p_{\text{remask}})$. The final transition is executed as follows:

$$x_s^{(i)} = \begin{cases} \tilde{x}^{(i)} & \text{if } x_t^{(i)} = [\text{mask}] \\ [\text{mask}] & \text{if } x_t^{(i)} \neq [\text{mask}] \text{ and } r_i = 1 \\ x_t^{(i)} & \text{O.W.} \end{cases}$$

Token Balance Verification:

Proposition: Under the proposed ECR scheme, the expected number of masked tokens at timestep s strictly equals $L * m_s$.

Proof:

Let L denote the total sequence length. At timestep t , the expected number of masked and unmasked tokens are $\mathbb{E}[|M|] = L * m_t$ and $\mathbb{E}[|U|] = L * (1 - m_t)$ respectively.

Each unmasked token is remasked independently with probability p_{remask} . The expected number of remasked tokens is $\mathbb{E}[R] = L * (1 - m_t) * p_{\text{remask}}$.

Each masked token is given an additional unmasking probability p_{extra} . The expected number of extra unmasked tokens is:

$$\mathbb{E}[E] = L * m_t * \left(p_{\text{remask}} * \frac{1 - m_t}{m_t} \right) = \mathbb{E}[R]$$

Consequently, the expected number of final masked tokens mathematically resolves to the target schedule:

$$\mathbb{E}[\text{Final Masks}] = \mathbb{E}[|M|] - L * (m_t - m_s) - \mathbb{E}[E] + \mathbb{E}[R] = L * m_s$$

4. Experiments

4.1 Experimental Setup

- **Model Details:** We used the model provided in the MDLM paper with a context length of 1024. The model contains approximately 130 million parameters and was trained using a forward diffusion process over one million steps on the OpenWebText corpus, processing roughly 33 billion tokens.
- **Inference Details:** Evaluations were executed over 1,024 diffusion steps using 100 examples (totaling 102,400 tokens) per configuration on an NVIDIA RTX 4090.
- **Datasets & Tasks:** We evaluated our approach on unconditional text generation to assess general language modeling quality, and on the CommonGen dataset to test the model’s reasoning ability in logically combining distinct concepts into coherent sentences.
- **Metrics:** Generative Perplexity (GenPPL) was evaluated using a pre-trained GPT-2 Large model to measure fluency. To assess diversity, we computed Self BLEU-4 (where a lower score indicates higher diversity among generated outputs) and Distinct- n metrics (the proportion of unique n -grams).

4.2 Main Results

Method	GenPPL ↓	Self BLEU-4 ↓	Distinct-2 ↑	Distinct-3 ↑
DDPM	25.7	0.056	0.79	0.97
CWS ($\beta = 1.5$)	22.6	0.064	0.79	0.97
ECR ($\gamma = 0.1$)	23.1	0.047	0.80	0.97
CWS + ECR	20.2	0.055	0.81	0.98

Table 1. Main Results on Generation Quality and Diversity

As shown in Table 1, both proposed samplers outperform the baseline DDPM in terms of fluency (lower GenPPL) while maintaining strong diversity metrics. Notably, combining CWS and ECR yields improvements over the baseline DDPM across all metrics, with a particularly large gain in GenPPL, which decreases from 25.7 to 20.2 .

4.3 CommonGen Results: Concept-to-Sentence Generation

In this experiment, for each sentence we provided the model with three randomly selected words (concepts), and the model was required to generate a grammatically correct sentence containing all of them, with a maximum length of 64 tokens. Since the model is not inherently prompt-based, the standard prefix-prompting approach performed poorly, resulting in a distribution shift due to a mismatch between training and inference conditions.

To address this issue, we adopted a constraint-based conditioning approach by placing the input tokens directly into masked positions at uniform intervals and allowing the model to generate the remaining tokens. This approach produced more coherent and higher-quality sentences.

Method	GenPPL ↓	Self BLEU-4 ↓	Distinct-2 ↑	Distinct-3 ↑
DDPM	145.7	0.183	0.65	0.83
CWS ($\beta = 1.5$)	66.4	0.215	0.73	0.89
ECR ($\gamma = 0.1$)	82.5	0.158	0.69	0.88
CWS + ECR	54.9	0.179	0.76	0.91

Table 2. Evaluation results on the CommonGen constrained generation task

As shown in Table 2, both proposed samplers significantly outperform the baseline DDPM in terms of fluency (lower GenPPL) on the constrained generation task. While CWS alone improves vocabulary diversity (higher Distinct-2 and Distinct-3) at the cost of structural similarity (higher Self-BLEU-4), ECR effectively mitigates this structural overlap. Notably, combining CWS and ECR yields improvements over the baseline DDPM across all metrics, with a particularly large gain in GenPPL, which dramatically decreases from 145.7 to 54.9 .

5. Ablation Study

We ablated the hyper-parameters β (confidence exponent) and γ (correction factor) to thoroughly understand model sensitivity.

CWS Method	GenPPL ↓	Self BLEU-4 ↓	Distinct-2 ↑	Distinct-3 ↑
$\beta = 0.5$	23.8	0.058	0.80	0.98
$\beta = 1.0$	23.5	0.060	0.80	0.97
$\beta = 1.5$	22.6	0.064	0.79	0.97
$\beta = 2.0$	26.4	0.035	0.81	0.98

Table 3. Ablation on Confidence Exponent (β)

As shown in Table 3, increasing β improves fluency up to an optimal threshold of 1.5 . However, an overly aggressive confidence bias ($\beta = 2.0$) forces premature commitments to suboptimal tokens, deteriorating GenPPL to 26.4 .

ECR Method	GenPPL ↓	Self BLEU-4 ↓	Distinct-2 ↑	Distinct-3 ↑
$\gamma = 0.05$	23.7	0.051	0.81	0.97
$\gamma = 0.10$	23.1	0.047	0.80	0.97
$\gamma = 0.20$	23.9	0.042	0.80	0.98
$\gamma = 0.50$	24.5	0.039	0.81	0.98

Table 4. Ablation on Correction Factor (γ)

For ECR (Table 4), a modest correction factor ($\gamma = 0.1$) achieves the optimal balance, yielding the highest overall fluency (GenPPL of 23.1) while maintaining strong generation diversity before the quality-diversity trade-off degrades fluency at higher γ values.

6. Conclusion

Ultimately, this project establishes that smarter inference-time sampling can dramatically enhance the capabilities of Masked Diffusion Language Models, requiring zero additional training or architectural changes. By implementing Confidence-Weighted Sampling to prioritize highly certain tokens, alongside Error-Corrective Resampling to iteratively refine mistakes, the proposed methods successfully reduce error propagation while strictly maintaining the required diffusion schedule. These compute-efficient strategies result in marked improvements across fluency, lexical diversity, and commonsense reasoning, confirming that refined sampling is a critical lever for advancing diffusion-based text generation.

References

1. Sahoo, S., et al. *Simple and Effective Masked Diffusion Language Models*. ICLR 2024.
2. Austin, J., Johnson, D. D., Ho, J., Tarlow, D., and van den Berg, R. *Structured denoising diffusion models in discrete state-spaces*. In Advances in Neural Information Processing Systems, 2021.
3. Lin, B. Y., et al. *CommonGen: A Constrained Text Generation Challenge for Generative Commonsense Reasoning*. In Findings of EMNLP, 2020.